

Nathaniel Schultz

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Education

- **University of Oklahoma** **Norman, OK**
B.S. in Electrical Engineering, Minor in CS | GPA: 3.88/4.0 *Expected Graduation: May 2026*
M.S. in Electrical Engineering (Accelerated) *Expected Graduation: May 2027*
 - *Selected Graduate Coursework:* Radar Imaging, Digital Signal Processing (*in progress*), Random Signals (*in progress*), Electromagnetic Fields and Wave Propagation (*in progress*), Fundamentals of Machine Learning (*Spring 2026*)
 - **Rose State College** **Midwest City, OK**
A.A.S in Cyber Security, GPA: 3.9/4.0 *July 2022*
 - Received CompTIA Network+ Certification
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Experience

- **Advanced Radar Research Center** **Norman, OK**
Undergraduate Research Assistant *Feb. 2024 – Present*
 - Developing software and algorithms for integrated sensing and communications (ISAC) research, focused on validating implementations through experimental testing and enabling subsequent algorithm refinement.
 - Researching the 5G New Radio (NR) protocol for future projects in determining the impact of the protocol in ISAC systems.
 - Participating in the 25-26 AFRL Software-Defined Radio (SDR) challenge, focusing on the implementation of a Multiple-Input Multiple-Output (MIMO) ISAC radar with tracking capabilities using the USRP-2901 SDR platform.
 - Conducted experimental research into the use of ultrawideband (UWB) radar techniques in biomedical imaging radar applications.
 - Investigated biological phantoms for use in a biomedical imaging radar, including dielectric property characterization and assessment of 3D-printed structures for manufacturing viability and long-term stability.
 - **University of Oklahoma IT** **Norman, OK**
IT Support Specialist & Mentor *Sept. 2023 – May 2024*
 - Assisted customers with various IT requests over the phone, in-person, and through online ticketing software.
 - Resolved over 100 tickets of varying difficulty, including software and hardware requests.
 - As a mentor, ensured newer co-workers were assisted with more difficult requests, involved in training new employees, and ensured everyone remained on-track with their work.
 - **PCI Energy Solutions** **Norman, OK**
Cyber Security Intern *June 2022 – Mar. 2023*
 - Developed an automated Python/AWS tool integrating REST APIs to alert the security team of new CVE vulnerabilities.
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Leadership

- **IEEE Student Branch** **Norman, OK**
Secretary *May 2025 – Present*
 - Leading the process of obtaining a website for our branch and acquiring graduation regalia.
 - Keep track of event attendance and metrics, and ensure events are properly reported to IEEE and the university.
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Core Technical Skills

Programming & Analysis: MATLAB, L^AT_EX, C, Python, Verilog

Tools & Simulation: Intel Quartus/QuuestaSim, GNURadio, SolidWorks, Autodesk Fusion, Ansys HFSS, Multisim

Test & Measurement: Oscilloscope, Vector Network Analyzer (VNA), Spectrum Analyzer

Research & Prototyping: Experimental Validation, Data Analysis in MATLAB, Electromagnetic Simulation, FDM 3D Printing, Soldering, Circuit Design/Verification